Hybrid Machine-Learning / Molecular-Mechanics Energy Functions for Condensed-Phase Simulations: Training and Molecular Dynamics

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Abstract

We describe a hybrid machine-learning / molecular-mechanics (ML/MM) energy function for condensed-phase simulations in which intramolecular and short-range intermolecular interactions are evaluated with a neural-network potential, while intermolecular Lennard-Jones and Coulomb interactions are supplied by a CHARMM/CGenFF-compatible classical term that is switched on as a function of monomer center-of-mass separation. The same hybrid total energy is assembled during supervised training and during molecular dynamics (MD), so that the learned model is trained on the Hamiltonian later used for dynamics. Mechanical and electrostatic embeddings for the classical Coulomb term are controlled by a charge-mode switch. Periodic long-range electrostatics are provided by optional Ewald / particle-mesh Ewald solvers. We outline validation targets—bonded MM parity, NVE energy conservation, rigid dimer scans, liquid densities, and free-energy profiles from umbrella sampling and CHARMM adaptive umbrella dynamics (ADUMB)—and provide regenerable workflow hooks; nu-

merical panels that are not yet frozen appear as explicit placeholders. Implementation details and CLI examples are collected in the Supporting Information.

Contents

1 Introduction

Machine-learned interatomic potentials have transformed the accuracy–cost trade-off for molecular simulation, from early high-dimensional neural network representations[1,2] and Gaussian approximation potentials[3,4] to message-passing architectures such as SchNet,[5] PhysNet,[6] SpookyNet,[7] and related force fields trained to near-quantum fidelity.[8–11] Nonetheless, treating large condensed-phase systems entirely with neural networks remains expensive when long-range electrostatics and many solvent molecules must be retained.

Classical molecular-mechanics (MM) force fields such as CHARMM/CGenFF[12–14] remain efficient for intermolecular interactions at longer range, and particle-mesh Ewald (PME) methods provide scalable periodic Coulomb sums.[15–17] Hybrid schemes that combine a local high-accuracy model with an efficient environment have a long history in QM/MM chemistry[18–20] and motivate analogous ML/MM constructions: ML for intramolecular and short-range many-body interactions, MM for the intermolecular tail and optional long-range Coulomb solvers.[6,21]

A practical hybrid scheme must expose a single total energy both when fitting parameters and when integrating equations of motion. The MMML package[21] implements that requirement through a shared hybrid forward pass used by training and MD entry points. The remainder of this article defines the hybrid energy (Section 2), summarizes training and condensed-phase protocols (Sections 3–4), describes charge embeddings (Section 5), and presents the validation and application targets that the accompanying Snakemake workflows are designed to regenerate (Section 6). Software command inventories and YAML templates are deferred to the Supporting Information.

2 Hybrid ML/MM Energy Function

2.1 Monomer Partition and Center-of-Mass Switching

Interactions are partitioned by monomer (typically one solvent residue). Let r_{IJ} denote the center-of-mass (COM) distance between monomers I and J. The hybrid potential takes the form

$$E = \sum_I E_I^{\mathrm{ML}} + \sum_{I<J} \big[s_{\mathrm{ML}}(r_{IJ})\, E_{IJ}^{\mathrm{ML}} + s_{\mathrm{MM}}(r_{IJ})\, E_{IJ}^{\mathrm{MM}}\big] + E^{\mathrm{LR}} + E^{\mathrm{wall/restr}}. \quad (1)$$

where E_I^{ML} is the intramolecular ML energy of monomer I, E_{IJ}^{ML} is the ML two-body (dimer) correction, and E_{IJ}^{MM} collects switched intermolecular Lennard-Jones and Coulomb contributions evaluated with CGenFF-compatible parameters. Optional long-range Coulomb beyond the real-space switch enters as E^{LR} (Ewald,[15] PME,[16,17] or related solvers). A short-range steric wall and restraints may contribute $E^{\mathrm{wall/restr}}$ (including optional Ziegler–Biersack–Littmark nuclear repulsion inside the neural model[22]).

For an isolated dimer AB, Equation 1 is equivalent to the training assembly

$$E = (1 - s)\,(E_A + E_B) + s\, E_{AB} + E_{\mathrm{MM}}, \quad (2)$$

with $s = s_{\mathrm{ML}}(r_{\mathrm{COM}})$, $\Delta E_{\mathrm{ML}} = E_{AB} - E_A - E_B$, and $E = E_A + E_B + s\,\Delta E_{\mathrm{ML}} + E_{\mathrm{MM}}$. The taper multiplies only the dimer interaction; monomer intramolecular energies remain fully on at all separations. Forces include the product-rule term arising from $\partial s/\partial \mathbf{R}$.

2.2 Default Handoff Window

With complementary handoff, $s_{\mathrm{ML}} + s_{\mathrm{MM}} = 1$ inside the ML↔MM transition. Production defaults used throughout training and MD YAML are listed in Table 1 and illustrated in

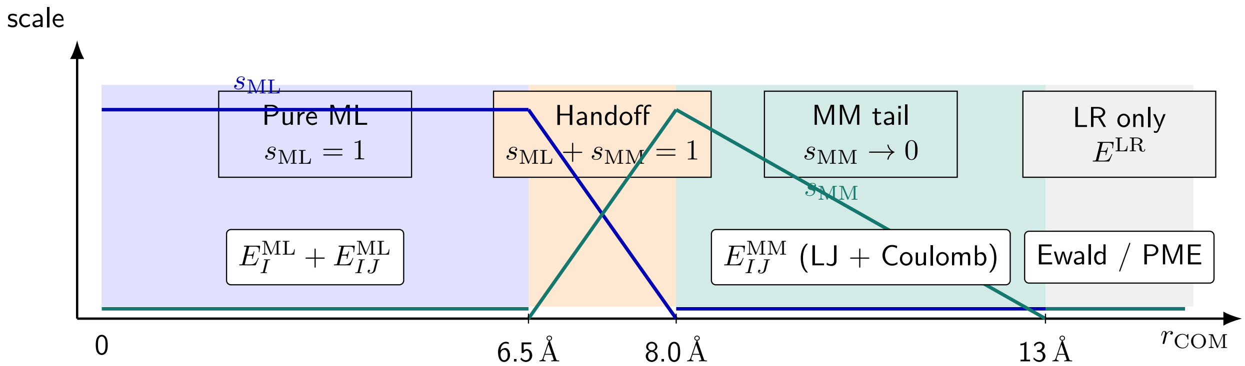

Figure 1: Schematic COM handoff for the hybrid energy in Equation 1. Below ≈ 6.5 Å the intermolecular ML dimer term is fully on; between 6.5 Å and 8.0 Å ML and MM scales are complementary; the switched MM pair term decays by 13 Å. Beyond the MM cutoff, only optional long-range Coulomb (E^{LR}) remains.

Figure 1.

Table 1: Default COM handoff parameters shared by hybrid training and MD.

Parameter	Symbol / role	Default
`mm_switch_on`	end of ML→MM handoff	8.0 Å
`ml_switch_width`	ML taper width	1.5 Å
`mm_switch_width`	MM outer tail	5.0 Å

2.3 Ownership of Terms

Table 2 summarizes which component owns each contribution.

3 Hybrid Training

Hybrid models are trained so that the supervised target is the hybrid total of Equation 2, not the bare neural energy. Datasets are covalent dimers with QM labels (R, Z, E, $\mathbf{F}$, optional multipoles)[24–27] augmented by CGenFF type indices and charges. The objective is a weighted sum of squared residuals on hybrid energies and forces (and optional dipole /

Table 2: Term ownership in the hybrid Hamiltonian.

Term	Owner	Notes
E_I^{ML}	PhysNet / Spooky / SO3LR[6,7,23]	Always on
E_{IJ}^{ML}	PhysNet / Spooky / SO3LR	COM-gated by s_{ML}
LJ + Coulomb pairs	CGenFF MM	Scaled by s_{MM}; q from charge mode
Optional LJ scales	Trainable $s^{\sigma}, s^{\varepsilon}$	Sidecar scales file
E^{LR}	Ewald / PME / ...	Long-range solver flag
Bonded CHARMM	Topology (Mode B)	Or jax-mm-spoof bonded clone

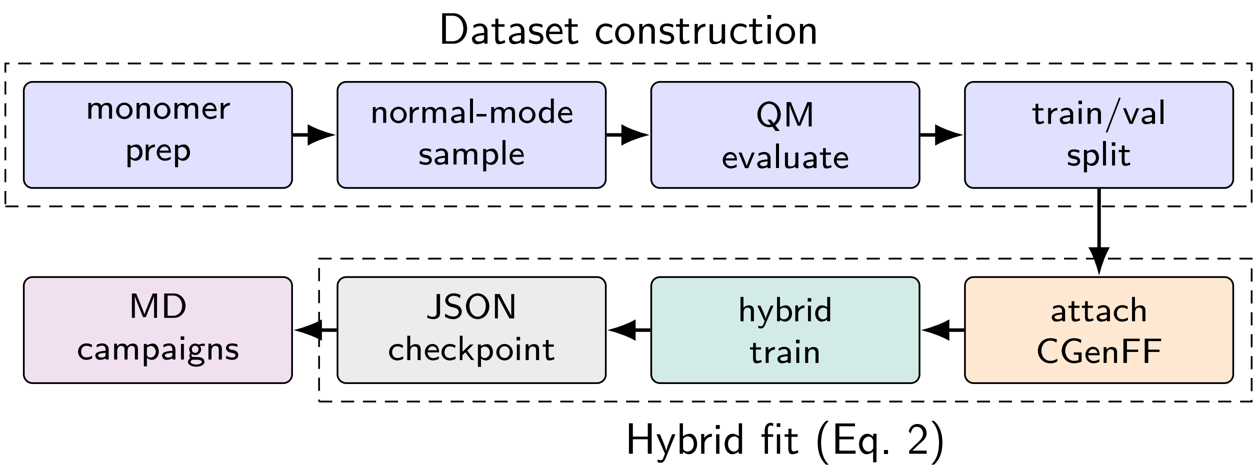

Figure 2: Hybrid training pipeline. Quantum labels on dimer geometries are augmented with CGenFF arrays; hybrid training fits Equation 2. The resulting portable JSON checkpoint is consumed unchanged by MD campaigns.

charge heads),

$$\mathcal{L} = w_E \|E - E^{\mathrm{ref}}\|^2 + w_F \|\mathbf{F} - \mathbf{F}^{\mathrm{ref}}\|^2 + w_D \|\boldsymbol{\mu} - \boldsymbol{\mu}^{\mathrm{ref}}\|^2 + w_q \|q - q^{\mathrm{ref}}\|^2. \tag{3}$$

Representative weights in the bundled fixed-charge template are $w_E = 1$, $w_F = 52.91$, $w_D = 27.21$, and $w_q = 14.39$. The PhysNet radial cutoff must be at least the MM handoff onset (Table 1). Portable JSON checkpoints (and optional LJ-scale sidecars) are written for MD deployment. The dataset → fit → MD pipeline is summarized in Figure 2; CLI flags and invocation examples are collected in Section S1.

4 Condensed-Phase Molecular Dynamics

4.1 Two Drivers, One Hamiltonian

Dynamics are organized as campaign YAML files that chain box preparation, minimization, heating, equilibration, and production under a shared set of handoff and long-range defaults (Supporting Information, Section S2).[28,29] Supported backends are PyCHARMM (CHARMM integrator with an MLpot USER callback),[12] jax-md (pure-JAX hybrid energy with a jax-md integrator),[30] and ASE for reference smokes. Thermostats and barostats follow standard Nosé–Hoover / MTTK practice.[31,32]

- Mode A (jax-md). CHARMM is used only to build the topology; every energy term is evaluated in a jitted hybrid energy function.
- Mode B (PyCHARMM). CHARMM advances the trajectory; the hybrid ML contribution enters each step through the MLpot callback while classical bonded / nonbonded bookkeeping follows the PSF.

Both drivers evaluate Equation 1 with the same handoff parameters and charge mode.

4.2 Box Preparation and Dynamics Stages

Dense liquids follow a two-phase protocol (Figure 3):

1. Phase A (MM certify). Packmol placement,[33] optional Monte Carlo density moves, and classical minimization / short dynamics until inter-monomer contact criteria pass.
2. Phase B (hybrid MD). Register the ML potential (or a jax-mm-spoof bonded clone for infrastructure tests), minimize under the hybrid energy, then heat $\rightarrow$ equilibrate $\rightarrow$ produce.

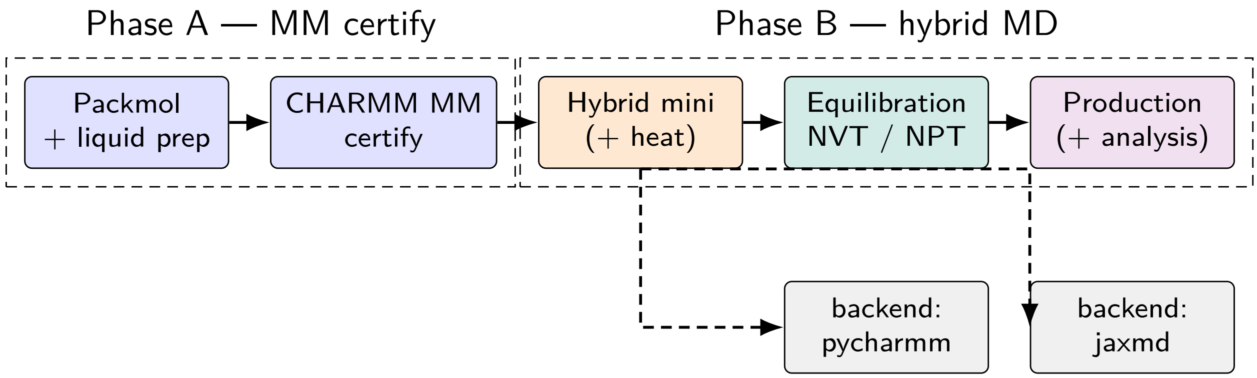

Figure 3: Condensed-phase campaign pipeline. Phase A builds and certifies a classical liquid box; Phase B registers the hybrid potential and runs mini/heat/equilibration/production on either the PyCHARMM or jax-md backend.

4.3 Free-Energy and Scanning Protocols

Beyond equilibrium liquid MD, the same hybrid energy supports:

- Rigid dimer / pair scans along a COM or atom–atom distance to validate the ML↔MM handoff against QM or DES reference curves;
- Umbrella sampling with harmonic restraints $W_k = \frac{1}{2}k_k(\xi - \xi_{0,k})^2$ and MBAR analysis[34,35] (gas-phase packed ML windows or hybrid mechanical embedding in explicit solvent);
- CHARMM ADUMB adaptive umbrella / WHAM paths[36,37] driven through PyCHARMM reaction-coordinate lingo for reactive complexes (e.g. Menshutkin-type $NH_3 + CH_3Cl$ studies[38,39]).

CLI entry points and example YAML are given in Sections S2 and S3.

5 Mechanical and Electrostatic Embeddings

The classical Coulomb term inside E_{MM} obtains its per-atom charges from a charge-mode switch. This choice is orthogonal to whether the neural model predicts charges for use inside E^{ML}.

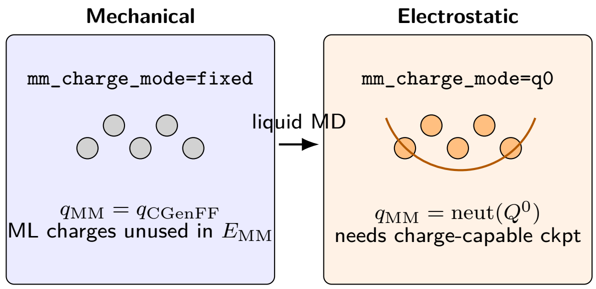

Figure 4: Liquid-capable embeddings for hybrid MM Coulomb. Mechanical embedding keeps fixed CGenFF charges in E_{MM}. Electrostatic embedding (Q^0) rebuilds neutralized monomer ML charges each evaluation.

Mechanical embedding (**`fixed`**). $q_{\mathrm{MM}} = q_{\mathrm{CGenFF}}$. Default for charge-less PhysNet checkpoints and for LJ-fitting stages.

Electrostatic embedding (**`q0`** / Q^0). $q_{\mathrm{MM}} = \mathrm{neutralize}(Q^0)$, where Q^0 denotes unperturbed monomer charge-head evaluations. Requires a charge-capable checkpoint.

Dimer-only modes (**`latent`**/Q^1). Partner-perturbed charges from the AB forward; undefined for $N > 2$ monomers and therefore not used in liquid production matrices.

Figure 4 contrasts the two liquid-capable embeddings.

6 Results and Illustrative Applications

Figures and tables below are drawn from regenerable artifacts under `artifacts/`, `results/`, and the simulation-robustness report assets (paths in Section S4 and `figures/extracted_metrics.json`). Panels still awaiting production $\langle\rho\rangle$ export remain marked (PLACEHOLDER).

6.1 Bonded MM Parity (jax-mm-spoof)

The hybrid jax-md path can replace the neural model by a JAX CGenFF bonded clone (`jax_mm_spoof`) so that drivers can be certified without a trained checkpoint. Bonded energy components for dichloromethane (DCM) and acetone (ACO) monomers match native PyCHARMM to machine precision (Table 3; `artifacts/jaxmd_cgenff_spoof_smoke/charmm_compare/`).

Table 3: Bonded energy parity: jax-mm-spoof versus native CHARMM (kcal mol^{-1}).

Case	E_{jax}	E_{CHARMM}	ΔE
DCM fixture	311.3070436736117	311.30704367361164	$+5.7 \times 10^{-14}$
DCM smoke-min	1.6671786898619212	1.667178689861924	-2.9×10^{-15}
ACO fixture	7.779347176527917	7.779347176527915	$+1.8 \times 10^{-15}$
ACO smoke-min	1.9927947535063575	1.9927947535063568	$+6.7 \times 10^{-16}$

6.2 Hybrid Training with Learnable LJ Scales

A 500-epoch hybrid `physnet-train` run with `learn_mm_lj_scales` on the Menshutkin (NH_3/CH_3Cl) enriched dataset (`examples/lj_scales/`, Orbax run `hybrid_mm_fixed_lj_scales-f7be8ce` yields the loss and MAE curves in Figure 5. Best validation loss is 5.243×10^{-3} at epoch 498; final validation energy / force MAEs are 3.05×10^{-3} eV ($\approx$ 0.070 kcal mol^{-1}) and 2.33×10^{-3} eV $^{-1}$ ($\approx$ 0.054 kcal mol^{-1} $^{-1}$). Dipole MAEs converge near 2.08×10^{-3} (Supporting Information).

Of 164 CGenFF types, three receive nonzero LJ-scale updates (Figure 6): HGA2, CG321, and CLGA1. The written sidecar places several of these scales outside the configured training bounds ($s^{\sigma} \in [0.95, 1.05]$, $s^{\varepsilon} \in [0.25, 4]$), so deployment should treat the sidecar as diagnostic until bounds / parameterization are revisited. Portable JSON + `hybrid_mm.json` live under `artifacts/nh3_ch3cl/ckpts/`; per-epoch metrics are archived as `figures/hybrid_mm_lj_metrics.csv`.

Table 4: Final hybrid + LJ-scale training metrics (epoch 500 / best valid).

Quantity	Value	Notes
Best valid loss	5.243×10^{-3}	epoch 498
Final valid loss	5.255×10^{-3}	epoch 500
Valid energy MAE	0.070 kcal mol^{-1}	3.05×10^{-3} eV
Valid force MAE	0.054 kcal mol^{-1} Å^{-1}	2.33×10^{-3} eV Å^{-1}
Types with $\lvert s-1 \rvert > 10^{-3}$	3 / 164	HGA2, CG321, CLGA1

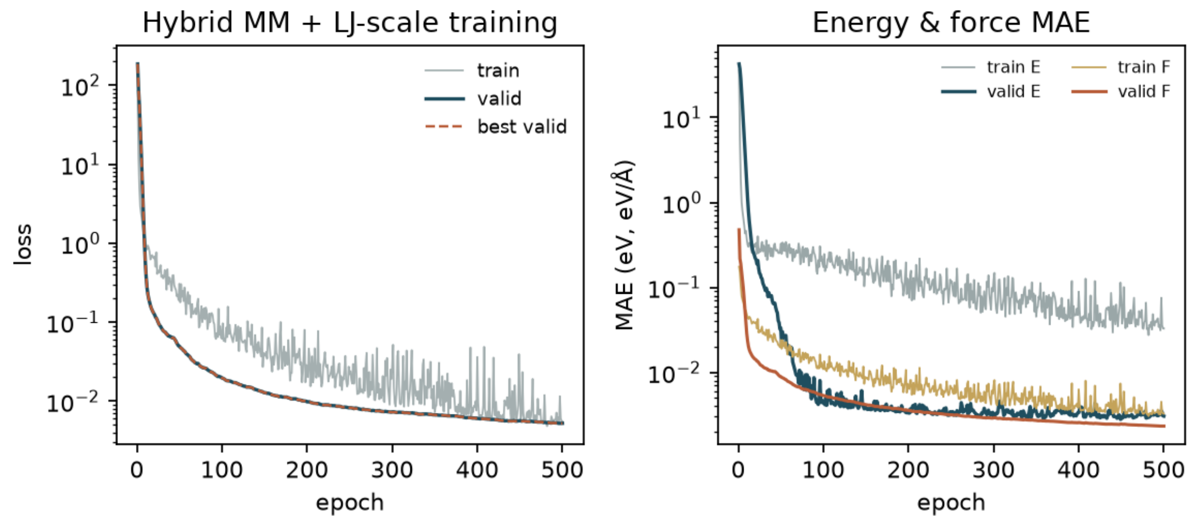

Figure 5: Hybrid MM training with learnable LJ scales (500 epochs; Menshutkin enriched NPZ). Left: train / valid / best-valid loss. Right: energy and force MAE. Metrics restored from Orbax epoch directories via OCDBT/zarr.

6.3 Energy Conservation (NVE)

Integrator robustness is illustrated by controlled NVE tests (Figure 7). For a small ethanol ASE NVE benchmark from the simulation-robustness report, total-energy fluctuations drop from $\mathrm{std}(E) = 0.0052\,\mathrm{kcal\,mol^{-1}}$ at $\Delta t = 0.5\,\mathrm{fs}$ to $0.0002\,\mathrm{kcal\,mol^{-1}}$ at $\Delta t = 0.1\,\mathrm{fs}$ ($\sim 23\times$). Mixed water / peptide–water NVE sweeps (`mixed_calculator_sweep`) give water-baseline drifts of $+0.29$ and $-0.14\,\mathrm{eV}$ over 10^4 steps (std $\approx 0.10\,\mathrm{eV}$); a mis-specified `mixed_core_vdw` setting is catastrophically unstable and is retained only as a negative control (Figure 8).

Hybrid DCM COM-cutoff ranking (`dcm3_nve_cutoff_sweep`) has completed engineering CSV summaries but currently shows unphysically high $\langle T \rangle$ on several presets; those rows are not promoted as paper conservation metrics until re-frozen.

6.4 Rigid Dimer and Pair Scans

Rigid intermolecular scans from the dimer-scan campaign (`results/dimer_scan_campaign/mbd_checkpoi` provide cleaned interaction curves for DCM, acetone (ACE), and water (TIP3) pairs (Figure 9). Panels plot E_{int} from `scan_results_clean.csv` with MP2 lateral-offset families and

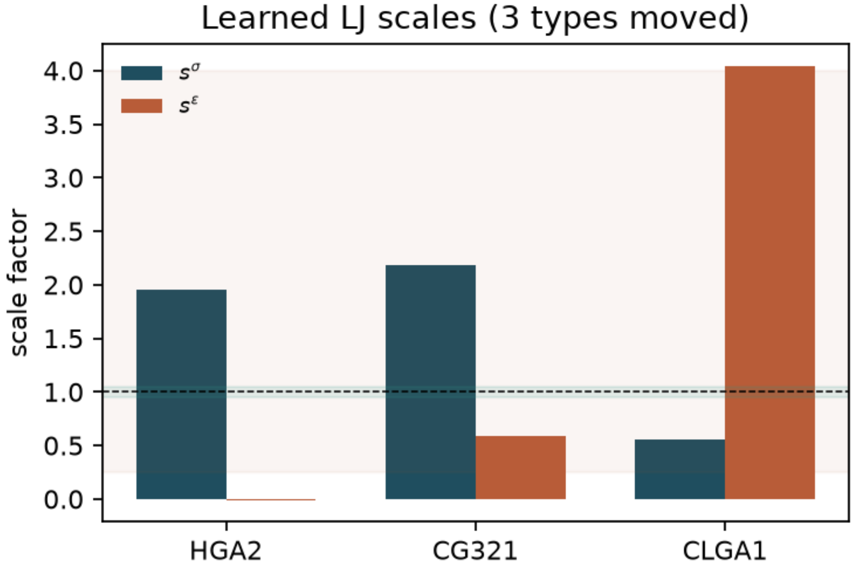

Figure 6: Learned MM LJ scale factors for the three CGenFF types that moved during training (HGA2, CG321, CLGA1). Dashed line: unscaled CGenFF ($s = 1$). Shaded bands mark the intended σ and ε bounds; several final scales lie outside those windows.

Table 5: Selected NVE conservation metrics from available artifacts.

System / setting	Source	Drift	Noise (std)	N_{steps}
Ethanol ASE, 0.1 fs	robustness report	—	0.0002 kcal mol^{-1}	short
Ethanol ASE, 0.5 fs	robustness report	—	0.0052 kcal mol^{-1}	short
Water baseline (seed 1)	mixed_calculator_sweep	+0.29 eV	0.095 eV	10^4
Water baseline (seed 2)	mixed_calculator_sweep	−0.14 eV	0.112 eV	10^4
DCM liquid NVE (paper preset)	dcm3_nve_cutoff_sweep	—	—	pending fr

bold MP2 / PBE0-D3(BJ) (plus GFN2-xTB on DCM–TIP3). Deepest electronic-structure wells on the highlighted $\Delta = 0$ rays are $\approx -2.38\,\mathrm{kcal\,mol^{-1}}$ (DCM–DCM, PBE0-D3(BJ) at $r = 4.50\,\text{Å}$) and $-2.31\,\mathrm{kcal\,mol^{-1}}$ (DCM–TIP3, PBE0-D3(BJ) at $r = 3.75\,\text{Å}$); MP2 wells are -1.17 and $-1.69\,\mathrm{kcal\,mol^{-1}}$, respectively. Short-range ACE QC geometries were largely rejected by cleaning, so ACE is kept in the numeric table (GFN2-xTB $\Delta = 2.5$) rather than the main figure. Oscillatory SpookyNet campaign overlays are omitted; hybrid ML/MM orientation scans are reported separately below.

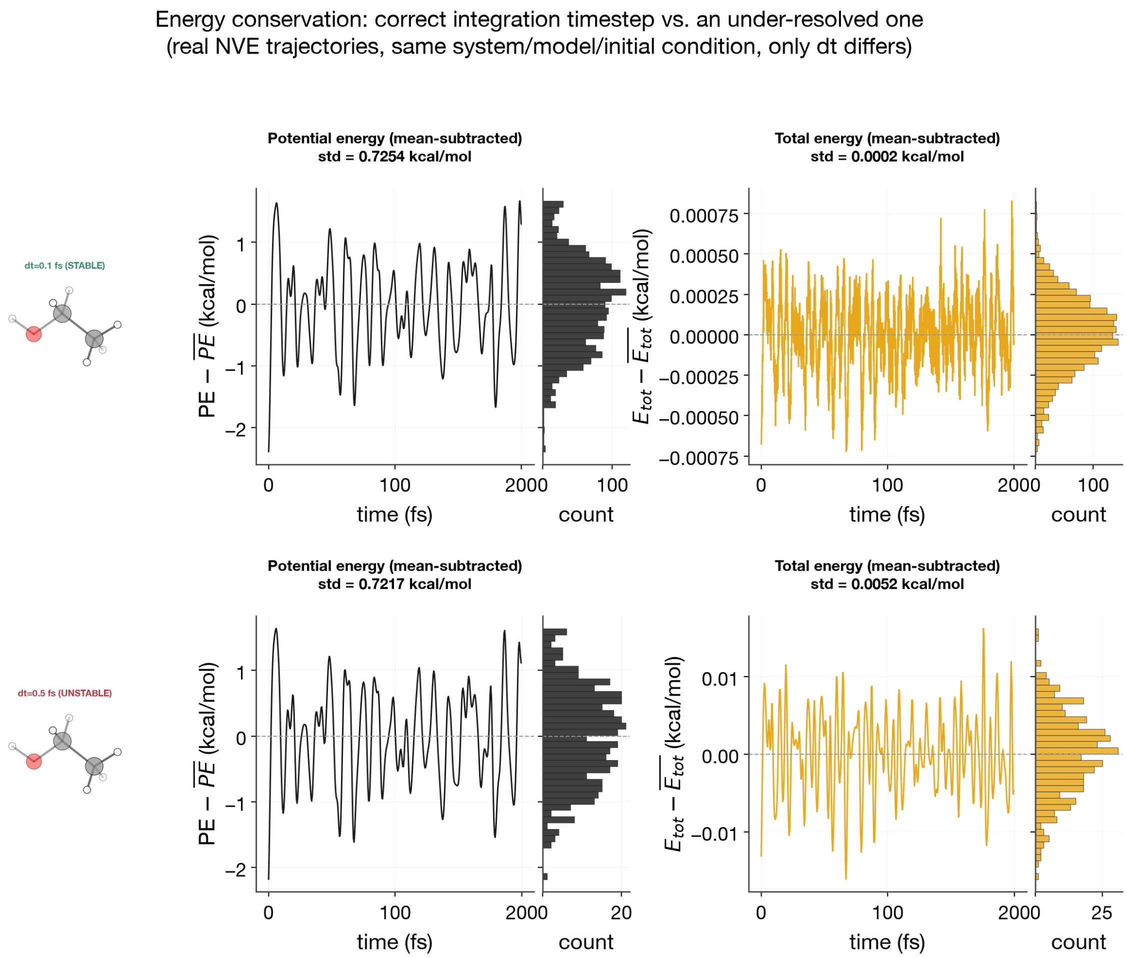

Figure 7: NVE total-energy conservation for a small ethanol system at two timesteps (simulation-robustness report). Tighter Δt suppresses high-frequency E_{tot} noise by more than an order of magnitude.

6.5 Hybrid Orientation Scans with Energy Components

To probe orientation dependence under the trained hybrid + learnable-LJ checkpoint, we scan DCM-like dimers ($Z = [6, 17, 17, 1, 1]$) over a Fibonacci S^2 direction set $\times$ super-Fibonacci SO(3) orientations ($6 \times 8 = 48$ rays) with COM separations $r \in [2.8, 12]$ Å (36 points). Energies and forces are evaluated with `hybrid_forward` using the portable PhysNet JSON and `hybrid_mm.json` LJ-scale sidecar from run `hybrid_mm_fixed_lj_scales-f7be8ce9-...`. Interaction energies are reported relative to the large-r asymptote; the ML piece is $E_{\text{ML}} = E_{\text{int}} - E_{\text{MM}}$ with E_{MM} the always-on CGenFF Coulomb/LJ (+wall) contribution from the hybrid assembly.

Table 6: Deepest cleaned QC / xTB interaction wells on the highlighted offsets.

Pair	Method	r / Å	Δ / Å	E_{int} / kcal mol^{-1}
DCM–DCM	PBE0-D3(BJ)	4.50	0.0	−2.38
DCM–DCM	MP2	4.50	0.0	−1.17
DCM–TIP3	PBE0-D3(BJ)	3.75	0.0	−2.31
DCM–TIP3	MP2	4.00	0.0	−1.69
ACE–ACE	GFN2-xTB	2.32	2.5	−3.25

Figure 10 shows the orientation envelope: contact geometries are strongly repulsive and orientation-sensitive, while the soft region ($r \gtrsim 3.5$ Å) yields a mean well of $\approx -1.58\,\mathrm{kcal\,mol^{-1}}$ and a deepest soft well of $-6.69\,\mathrm{kcal\,mol^{-1}}$ at $r = 4.38$ Å. Figure 11 decomposes total, ML, and MM interaction curves along representative rays (deep / median / shallow soft wells) together with the COM handoff s_{ML} (dotted/dashed verticals at 6.5 / 8.0 Å). Mean components over all rays (Figure 12) confirm that short-range binding is carried by the ML term inside $s_{\mathrm{ML}} = 1$, with a smooth decay of $\langle s_{\mathrm{ML}} \rangle$ through the handoff window. Because raw atomic $|F|$ retains intramolecular residuals even at large r, we report force changes relative to the separated asymptote, $|F - F_\infty|$, and the net monomer COM interaction force (Figure 13). Both metrics decay through the soft well and are already small entering the handoff window (median $\langle |F - F_\infty| \rangle \approx 2.6\,\mathrm{kcal\,mol^{-1}\,Å^{-1}}$ at $r = 6.5$ Å). CSV tables are archived as `figures/hybrid_orient_components.csv`.

The same portable checkpoint (run `hybrid_mm_fixed_lj_scales-32a09175-…`) applied to acetone dimers ($Z = [6, 6, 6, 8, 1 \times 6]$, CGenFF-enriched from `fixed-acetone-only_MP2`) is an out-of-distribution transfer test (Figure 14). Soft-region wells are shallow (mean $\approx -0.17\,\mathrm{kcal\,mol^{-1}}$, deepest $\approx -0.67\,\mathrm{kcal\,mol^{-1}}$ near the handoff), while $r \sim 5$ Å remains strongly repulsive across orientations. Force residuals relative to the separated asymptote stay large into the handoff window—consistent with a DCM-trained hybrid that has not seen ACE chemistry. ACE-specific hybrid training (or fine-tuning) is required before promoting acetone condensed-phase MD from this checkpoint.

6.6 Liquid Densities and Methane/Ewald Smokes

Dichloromethane and acetone liquids are targeted under NPT/NVT hybrid dynamics (`pbc_liquid_density`
Production artifact trees exist, but `density_g_cm3_mean` fields are still `null` in the available receipts, so Table 7 remains a reporting template. Engineering smokes for liquid methane with `lr_solver: ewald` and mechanical / electrostatic embeddings (DES and So3LR checkpoints) completed successfully (Table 8); density is an NVT input there, not an NPT observable.

Table 7: (PLACEHOLDER) Liquid densities under hybrid MD ($T = 300\,\mathrm{K}$ unless noted). $\langle\rho\rangle$ awaiting production metrics export.

Solvent	N	Ens.	Backend	Emb.	$\langle\rho\rangle$ ($\mathrm{g\,cm^{-3}}$)	Expt. ($\mathrm{g\,cm^{-3}}$)
DCM	—	NPT	jaxmd	mech.	—	∼1.33
DCM	—	NPT	pycharmm	mech.	—	∼1.33
ACO	—	NPT	jaxmd	m./es.	—	∼0.78
CH_4	8	NVT	both	m./es.	(imposed)	—

Table 8: Methane/Ewald embedding smoke matrix (`artifacts/pbc_methane_ewald_smoke_embeddings/`, $T = 100\,\mathrm{K}$, $L = 24\,\mathrm{Å}$, 8 methanes). Engineering PASS only—not liquid-property claims.

Checkpoint	Embedding	Backend	Result
DESdimers	mechanical (`fixed`)	pycharmm	PASS
DESdimers	mechanical	jaxmd	PASS
So3LR epoch0013	mechanical	pycharmm	PASS
So3LR epoch0013	mechanical	jaxmd	PASS
So3LR epoch0013	electrostatic (`q0`)	pycharmm	PASS
So3LR epoch0013	electrostatic	jaxmd	PASS

6.7 Umbrella Sampling and MBAR Free Energies

Gas-phase Menshutkin-type umbrella sampling with MBAR reweighting[34,35,38–41] is available from `artifacts/umbrella/umbrella_summary.json` (Figure 16). Twenty windows span $\xi = r(\mathrm{C{-}Cl}) - r(\mathrm{C{-}N}) \in [1.3, 3.5]\,\mathrm{Å}$ at 300 K with 2001 frames per window. The MBAR PMF barrier is $12.99 \pm 1.16\,\mathrm{kcal\,mol^{-1}}$ at $\xi \approx 2.34\,\mathrm{Å}$ (relative to the $\xi = 1.3\,\mathrm{Å}$ minimum). Explicit-solvent `hybrid_jaxmd` TIP3P windows[42] are present as partial runs without a finished MBAR block and are left for a later freeze.

Table 9: Umbrella / MBAR summary for the gas-phase reactive PMF.

Engine	N_{win}	Frames/win	ξ / Å	T / K	Barrier / kcal mol^{-1}
packed_ml + MBAR	20	2001	1.3–3.5	300	12.99 ± 1.16
hybrid_jaxmd (TIP3)	—	—	—	300	pending MBAR

6.8 CHARMM ADUMB Paths

Adaptive umbrella sampling in CHARMM (ADUMB)[12,36,37] is wired for hybrid PyCHARMM+MLpot Menshutkin setups (`examples/m/yaml/adumb_nc_distance.yaml`). The archived `adumb_nc_distance` job wrote `ADUMB-WUNI.DAT` and `UMBCOR` but exited with code 2 before a converged WHAM profile; coverage stays on only a few of the 40 bins (Figure 17). We therefore report the sampling-status figure rather than a barrier height. A completed ADUMB PMF suitable for overlay with Figure 16 remains a production target.

6.9 Simulation Matrix

Table 10 summarizes what is filled versus pending for the first manuscript freeze.

Table 10: Status of regenerable Results panels.

System	Protocol	Status	Primary source
DCM/ACO bonded	spoof parity	filled	`jaxmd_cgenff_spoof_smoke`
NH_3/CH_3Cl hybrid train	LJ scales + loss curves	filled	`artifacts/nh3_ch3cl/ckpt`
Ethanol / water NVE	conservation	filled	robustness + mixed_calculat
DCM/ACE/TIP3 pairs	rigid scans	filled	`dimer_scan_campaign`
DCM-like hybrid	orient. scans + components	filled	`hybrid_orient_*` figures
NH_3/CH_3Cl	umbrella+MBAR	filled (gas)	`artifacts/umbrella/`
NH_3/CH_3Cl	ADUMB	status only	`adumb_nc_distance`
CH_4 Ewald	embedding smokes	PASS (eng.)	`pbc_methane_ewald_smoke_`
DCM/ACO liquid	NPT $\langle\rho\rangle$	pending metrics	`pbc_liquid_density_dyn`

7 Conclusions

The hybrid ML/MM formulation in Equation 1 and the training assembly in Equation 2 provide a closed loop from dimer QM data to condensed-phase dynamics and free-energy protocols. Mechanical and electrostatic embeddings select how classical Coulomb charges are obtained without changing ML/MM ownership of bonded and short-range many-body terms. Shared handoff defaults and portable JSON checkpoints keep training and MD numerically aligned across the jax-md and PyCHARMM drivers. Filled Results panels already include bonded spoof parity, NVE conservation illustrations, rigid dimer scans, hybrid multi-orientation component/ force curves, and a gas-phase umbrella/MBAR PMF; liquid $\langle\rho\rangle$ and a converged ADUMB WHAM profile remain the principal outstanding production exports.

Acknowledgement

We thank members of the Meuwly group for discussions on hybrid ML/MM workflows. Computational resources at the University of Basel are gratefully acknowledged.

Supporting Information Available

Supporting Information is appended at the end of this file (CLI inventories, YAML templates, free-energy command examples, and workflow provenance notes). Extended CLI documentation also lives in the MMML documentation tree.

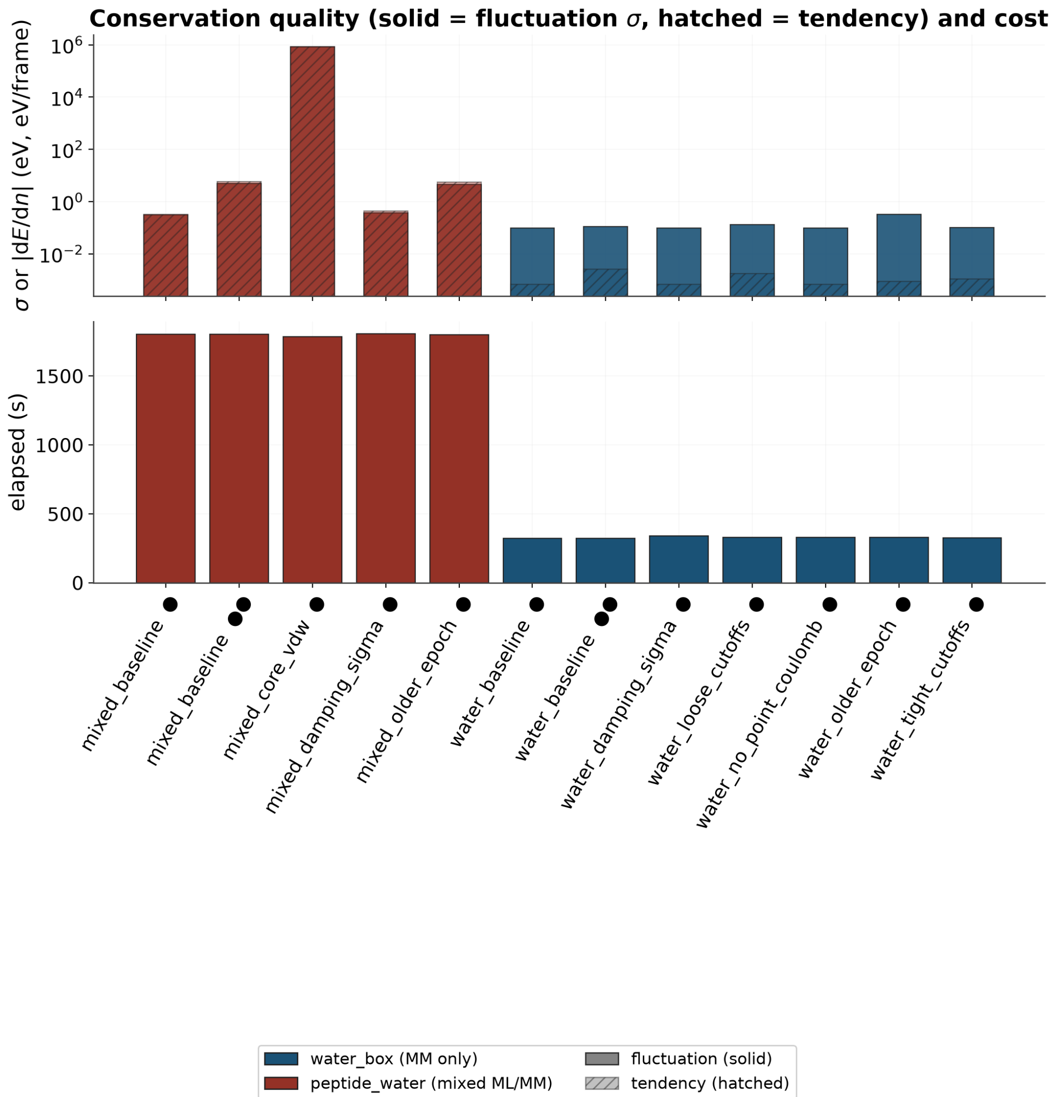

Figure 8: Mixed-calculator NVE sweep summary bars (water and peptide–water settings). Stable water baselines contrast with the exploding `mixed_core_vdw` control (`workflows/mixed_calculator_sweep/`).

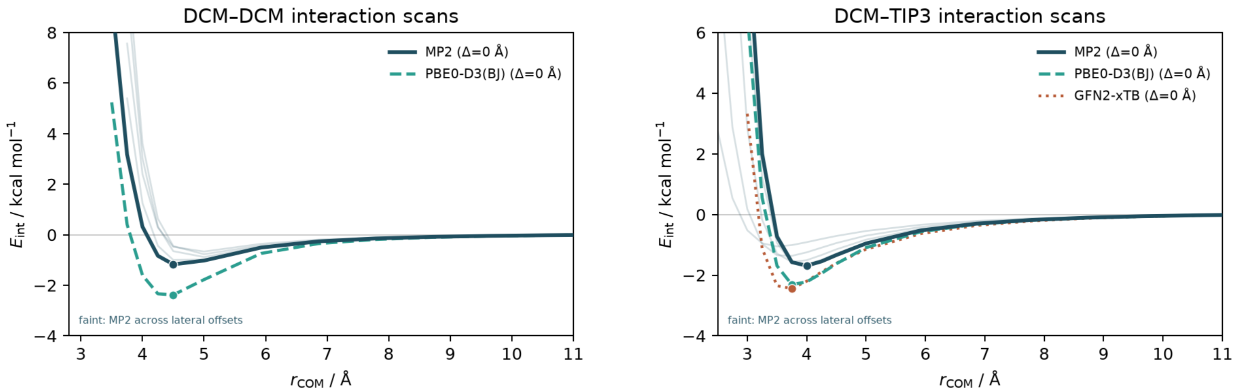

Figure 9: Rigid dimer interaction scans for DCM–DCM (left) and DCM–TIP3 (right) from the dimer-scan campaign (`scan_results_clean.csv`). Faint curves: MP2 across lateral offsets. Bold: MP2 and PBE0-D3(BJ) at $\Delta = 0$ (DCM–TIP3 also shows GFN2-xTB). Energies are cleaned interaction values.

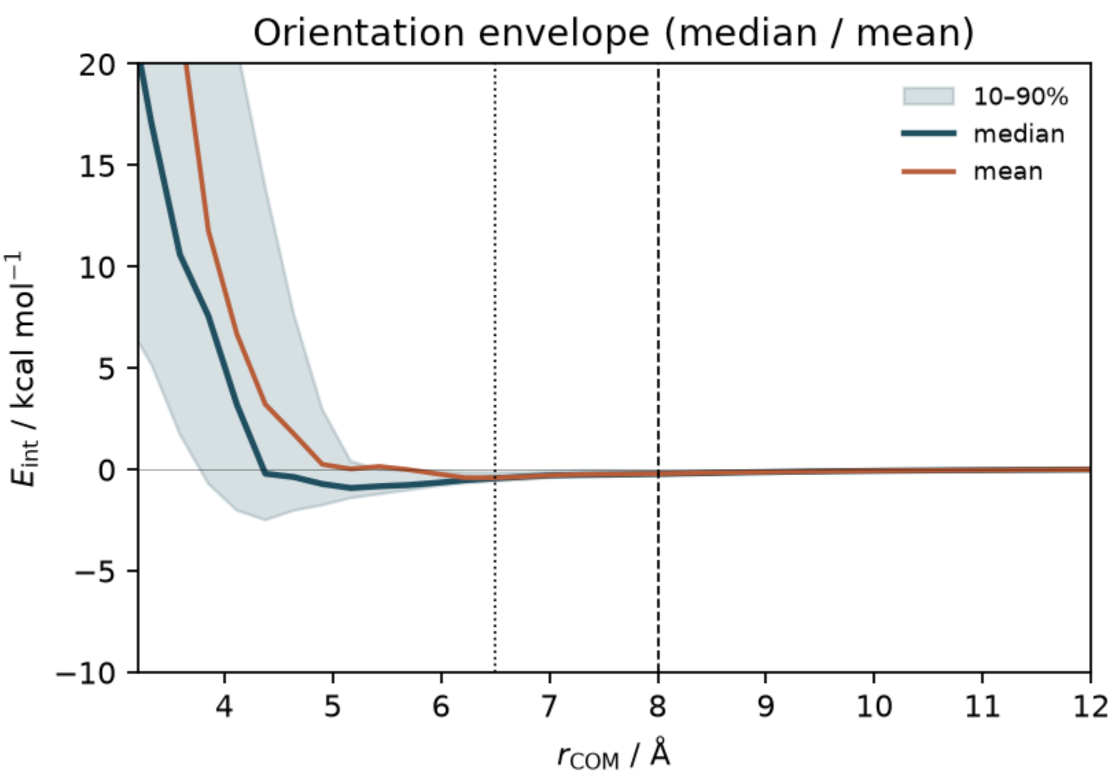

Figure 10: Hybrid DCM-like dimer interaction envelope over 48 relative orientations (median, mean, and 10–90% band). Vertical dotted/dashed lines mark the ML→MM handoff window (6.5–8.0 Å).

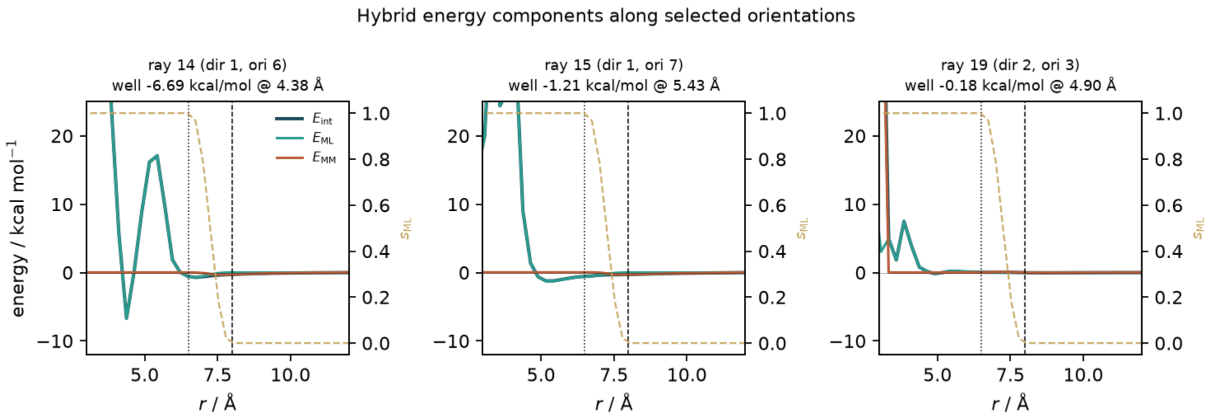

Figure 11: Hybrid energy components along three representative orientation rays (deepest, median, and shallowest soft-region wells). Solid curves: E_{int}, E_{ML}, and E_{MM} (kcal mol^{-1}); dashed gold: s_{ML}.

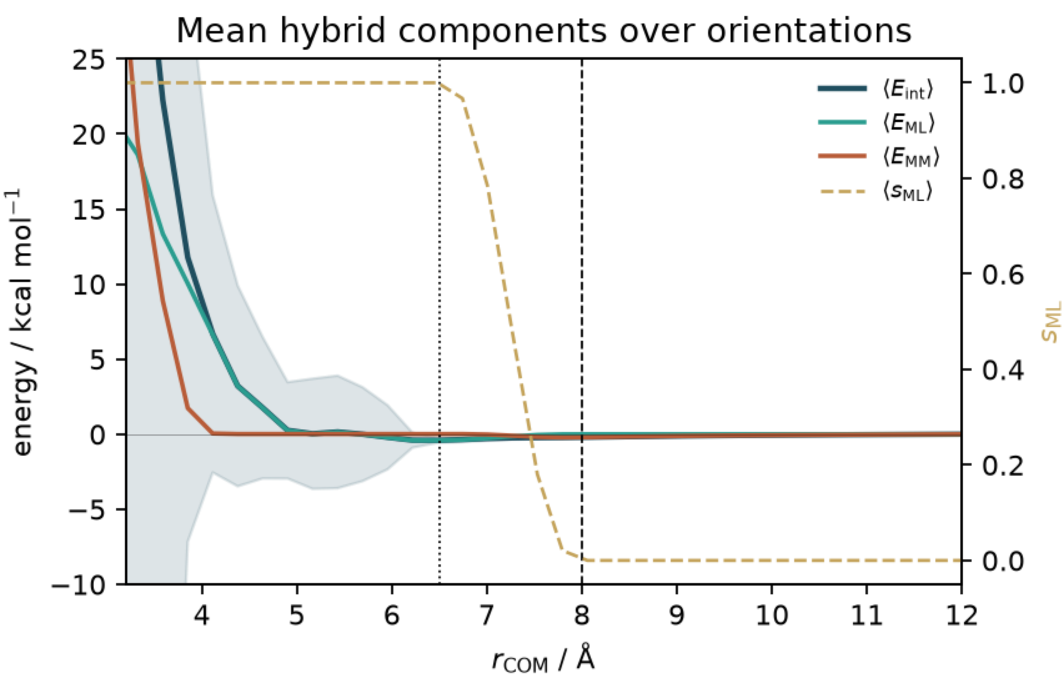

Figure 12: Mean hybrid components $\langle E_{\mathrm{int}} \rangle$, $\langle E_{\mathrm{ML}} \rangle$, $\langle E_{\mathrm{MM}} \rangle$ and $\langle s_{\mathrm{ML}} \rangle$ over the 48 orientation rays. Shaded band: $\pm 1\sigma$ on $\langle E_{\mathrm{int}} \rangle$.

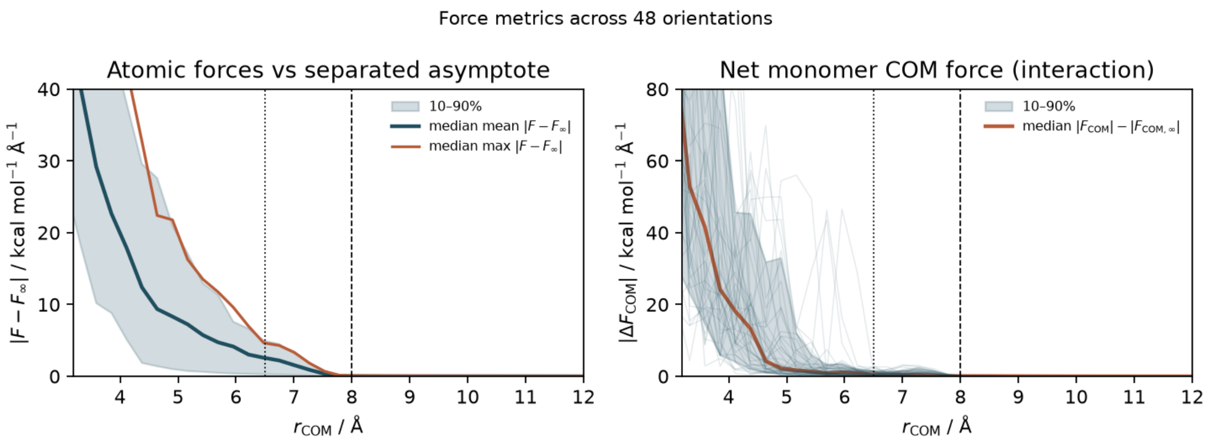

Figure 13: Force metrics across 48 orientations (kcal mol^{-1} Å^{-1}). Left: atomwise $|F - F_\infty|$ (median mean/max; 10–90% band). Right: net monomer COM interaction force relative to the separated limit. Vertical dotted/dashed lines mark the ML→MM handoff.

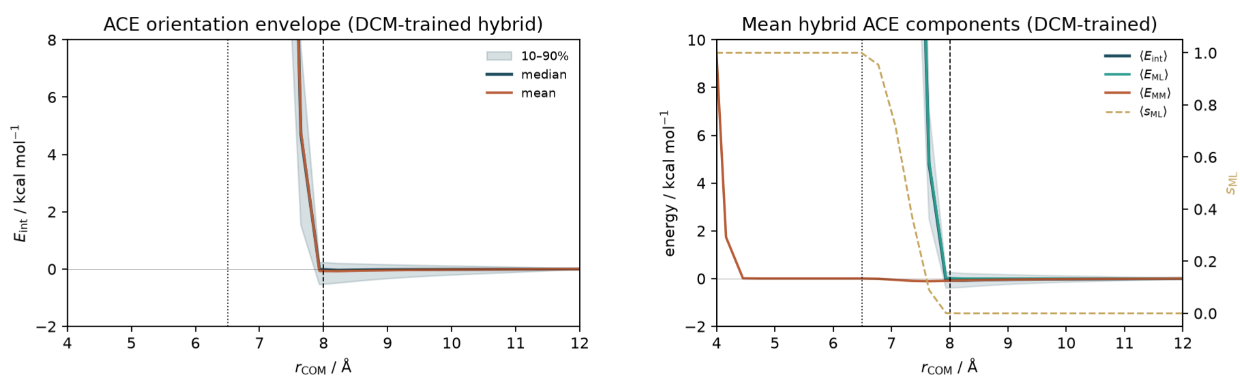

Figure 14: Acetone hybrid orientation scans with the DCM-trained LJ-scale checkpoint. Left: E_{int} envelope. Right: mean ML/MM components and s_{ML}. Soft wells are shallow; mid-range repulsion and large force residuals mark OOD transfer.

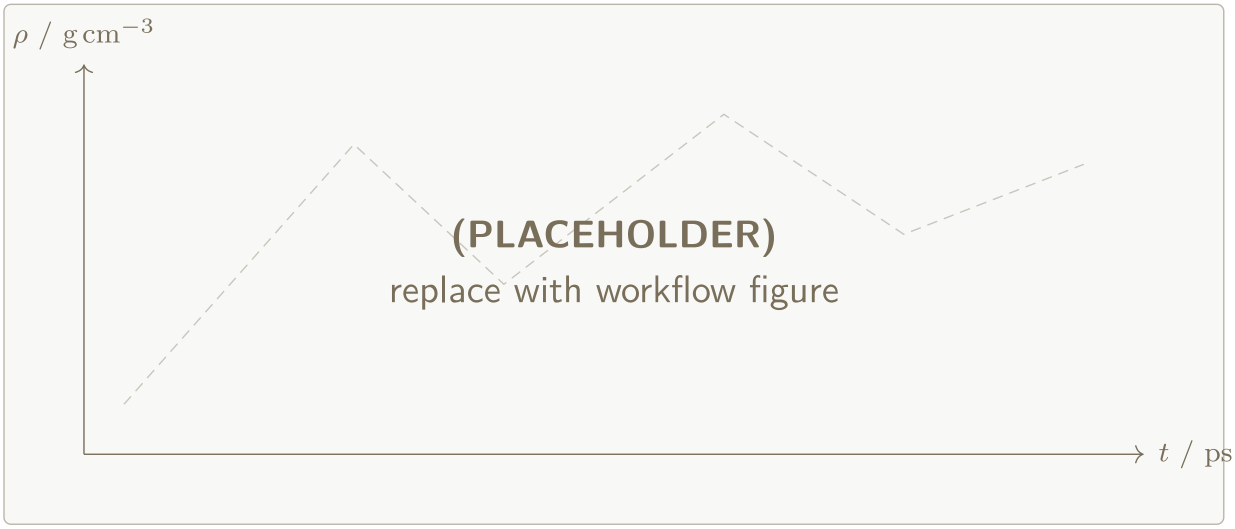

Figure 15: (PLACEHOLDER) Density time series $\rho(t)$ for hybrid DCM and ACO production windows once `metrics.json` density fields are populated by `pbc_liquid_density_dyn`.

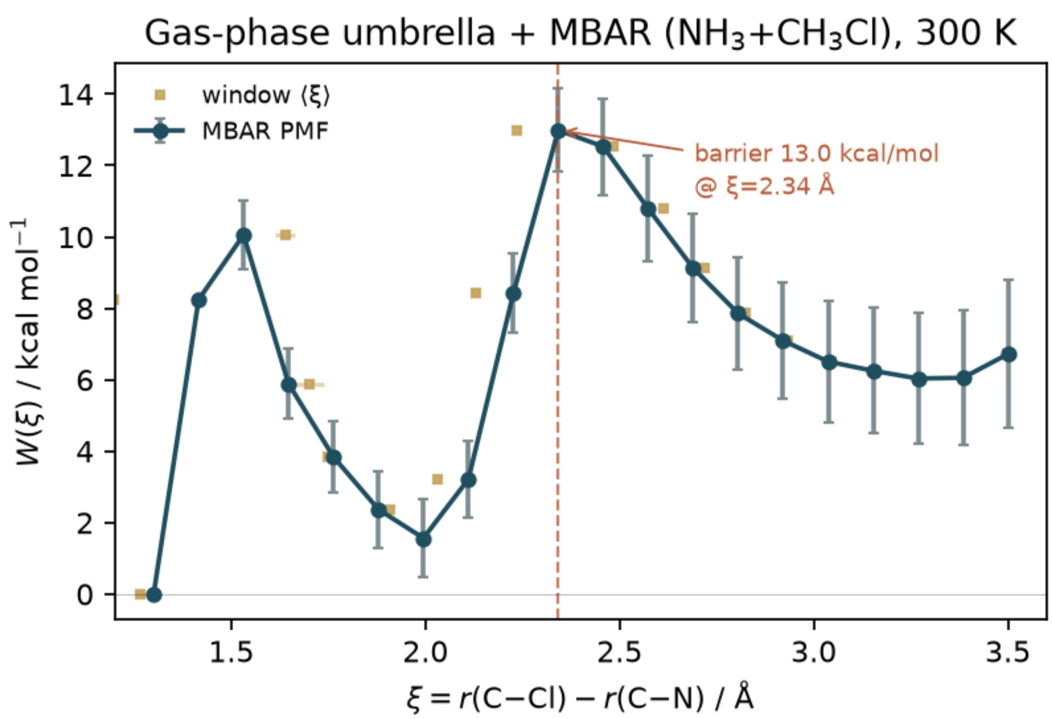

Figure 16: Gas-phase umbrella + MBAR PMF for NH_3+CH_3Cl along $\xi = r(\text{C}-\text{Cl}) - r(\text{C}-\text{N})$ at 300 K (20 windows; source `artifacts/umbrella/`).

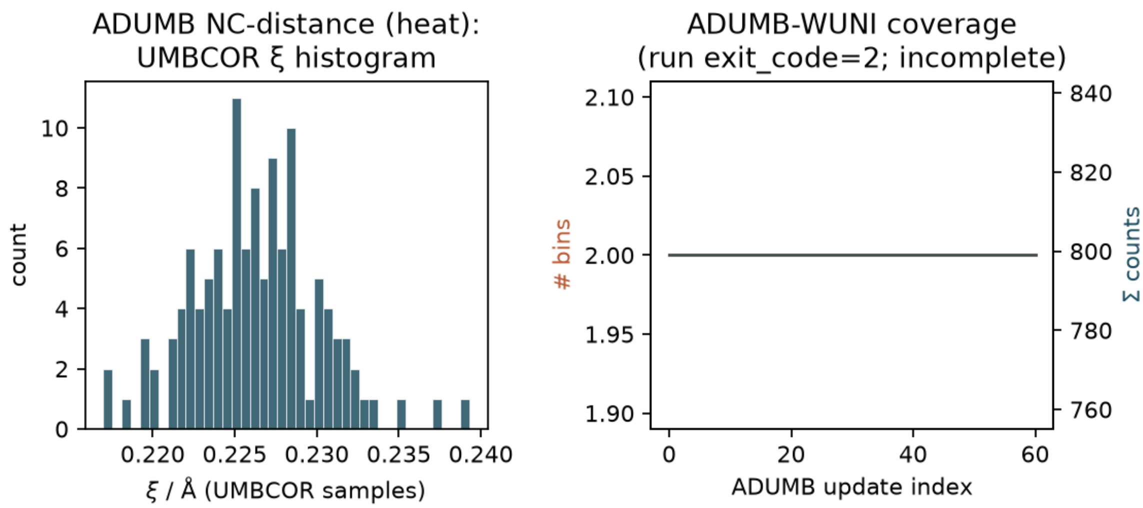

Figure 17: CHARMM ADUMB status for the NC-distance Menshutkin setup (`artifacts/nh3_ch3cl/adumb_nc_distance/`): UMBCOR ξ histogram during heat (left) and ADUMB-WUNI bin coverage versus update index (right). The run did not finish cleanly (`exit_code=2`); no WHAM barrier is claimed.

Supporting Information

Hybrid Machine-Learning / Molecular-Mechanics Energy Functions for Condensed-Phase Simulations

Eric Boittier and Markus Meuwly

S1 Training CLI and Flags

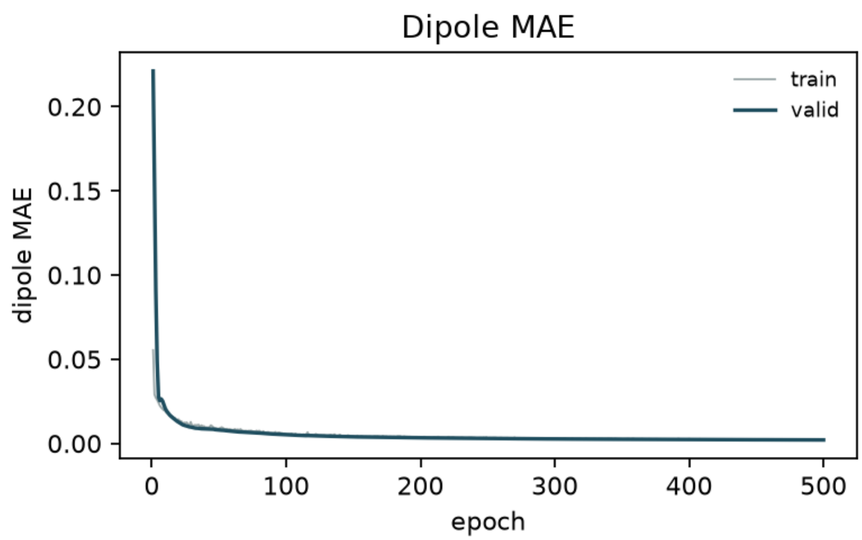

Figure S1: Dipole MAE versus epoch for the hybrid + LJ-scale Menshutkin run (companion to Figure 5).

Hybrid models are trained with

```
mmml physnet-train --hybrid-mm [--config train.yaml]
```

Essential flags (YAML or CLI):

- `--hybrid-mm` — enable hybrid total of Equation 2
- `--mm-charge-mode` {`fixed`, `q0`, `latent`, ...}

- `--charges` / `--no-charges` — neural charge head
- `--ml-switch-width`, `--mm-switch-on`, `--mm-switch-width`
- `--lr-solver` {`mic`, `ewald`, `nvalchemiops_pme`, ...}
- `--learn-mm-lj-scales` — optional trainable LJ scales

CGenFF fields in the NPZ (`cgenff_type_idx`, `mol_id`, `cgenff_charge`, master σ/ε) are prepared after electronic-structure evaluation of normal-mode samples, e.g.

```
mmml prepare-mm-dataset ...
mmml physnet-train --hybrid-mm --config train.yaml
```

S2 MD Campaign CLI and YAML

Dynamics are launched with

```
mmml md-system --config campaign.yaml --run-all [--resume]
```

A campaign file supplies global `defaults` (composition, box, checkpoint, cutoffs, `lr_solver`, `mm_charge_mode`, backend options) and an ordered list of `runs`/`jobs`. Dependencies are resolved automatically; each leg writes a handoff state consumed by the next stage.

S2.1 Minimal campaign sketch

```
defaults:
  composition: "DCM:64"
  backend: jaxmd
  setup: pbc_npt
  checkpoint: path/to/hybrid.json
  mm_switch_on: 8.0
  ml_switch_width: 1.5
```

```
  mm_switch_width: 5.0
  lr_solver: ewald
  mm_charge_mode: fixed
runs:
  - { id: mini, stage: minimize }
  - { id: heat, stage: heat, depends: [mini] }
  - { id: equi, stage: equilibrate, depends: [heat] }
  - { id: prod, stage: produce, depends: [equi] }
```

S2.2 Paper workflow entry points

- `workflows/pbc_liquid_density_dyn/` — DCM and ACO liquids
- `workflows/pbc_methane_ewald/` — methane + Ewald, embedding sweep
- `workflows/dcm3_nve_cutoff_sweep/` — NVE / cutoff ranking
- `workflows/jaxmd_cgenff_spoof_smoke/` — bonded parity
- `workflows/des_dimer_pair_scans/` — rigid DES panels

Table S1: Software stack for hybrid training and MD.

Component	Role
MMML[21]	CLI, hybrid calculator, workflows
JAX / jax-md[30]	Differentiable hybrid energy; jaxmd backend
PyCHARMM / CHARMM[12]	Topology, classical MM, pycharmm backend
PhysNet / SpookyNet / SO3LR[6,7,23]	Neural short-range models
Snakemake (+ Slurm)	Parameter sweeps and cluster submission

S3 Free-Energy Protocols (Umbrella and ADUMB)

S3.1 Batched / hybrid umbrella sampling

Gas-phase packed windows:

```
mmml umbrella-sample \
  --checkpoint model.json \
  --structure seed.xyz \
  --config umbrella.yaml
mmml umbrella-mbar --input umbrella_windows.json -o pmf/
```

Explicit-solvent mechanical embedding uses `engine: hybrid_jaxmd` with a PSF/PDB/box and an ML region (reactive complex) while solvent remains MM. Snapshots store unbiased hybrid energies so MBAR does not need to rebuild the Hamiltonian.

S3.2 CHARMM ADUMB

ADUMB is enabled via PyCHARMM pre-dynamics lingo (reaction-coordinate umbrella with `ADUMBRXNCOR` in the CHARMM build). Typical artifacts: `ADUMB-WUNI.DAT`, `UMBCOR`, `RXNCOR_TRACE.DAT`. NAME tokens are ≤ 4 characters. Example trees live under `examples/m/` and the `nh3_ch3cl` documentation.

Table S2: (PLACEHOLDER) Extended free-energy provenance (SI).

Protocol	CV	Artifact root	Git SHA
umbrella+MBAR	ξ	`artifacts/.../pmf/`	—
ADUMB+WHAM	ξ / multi	`artifacts/.../adumb/`	—

S4 Workflow Provenance

Each directory that feeds a main-text figure or table should contain:

1. frozen inputs (`request.json` / job YAML);
2. `metrics.json` (numbers that enter the paper);
3. plot PNG/PDF;
4. `provenance.json` — git SHA, checkpoint hash, host/partition;
5. optional `proof.json` when under validation-campaign gates.

Recommended manuscript aggregator: `workflows/manuscript_hybrid_liquids/` writing artifacts/manuscript_hybrid_liquids/<fig_or_table_id>/.

Table S3: Figure / table ↔ artifact map (main text).

Panel	Content	Source
Fig. 5	Hybrid train curves	`artifacts/nh3_ch3cl/ckpts/…f7be8ce9…`
Fig. 6	LJ scales	same run `hybrid_mm.json`
Fig. 7	Ethanol NVE	`docs/robustness-report-assets/`
Fig. 8	Mixed NVE bars	`mixed_calculator_sweep/results/`
Fig. 9	DCM/ACE scans	`results/dimer_scan_campaign/`
Fig. 16	MBAR PMF	`artifacts/umbrella/`
Fig. 17	ADUMB status	`artifacts/nh3_ch3cl/adumb_nc_distance/`
Fig. 15	$\rho(t)$	pending `pbc_liquid_density_dyn`
Table 3	Spoof parity	`jaxmd_cgenff_spoof_smoke`
Table 8	CH_4 embeddings	`pbc_methane_ewald_smoke_embeddings`

Copied PNG/PDF panels and a machine-readable summary live under `latex/figures/` (`extracted_metrics.json`).

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